DSAI5101 Statistical Data Mining

# Midterm Report Guidelines

Cat Image Arena Dataset

**Report and code due: 12:00 noon, Monday, 2 November 2026**

(Hong Kong time)

## 1. Project objective and choice of problem

Each group will conduct a data-analysis project using the Cat Image Arena Dataset. **There is no single prescribed research question.** Your group must formulate a clear, meaningful question and explain why it is worth studying. The quality, relevance, and ambition of the problem you choose will contribute to your final midterm score, together with how well you investigate it.

You may focus on exploration and interpretation, prediction, or a combination of both. Possible directions include disagreement in image preferences, differences between comparisons within the same cat and between different cats, or prediction of the preferred image. These are examples, not required topics, please do not be limited in these examples. **Prediction is optional.** A well-designed explanatory or exploratory study can receive a strong score.

## 2. Understanding the dataset

See Readme file.

## 3. Report contents and assessment criteria

**The report may be submitted as PowerPoint (.ppt or .pptx) or PDF and will be used directly for the group presentation.** Organise it as a presentation-ready account of the question, analysis, evidence, and conclusions. The submitted report is the main basis for assessing the project and should include the following elements.

1. **Title and group information.** State the project title, group number, and each member's name.
2. **Research question and motivation.** Define the problem precisely, explain its importance, and state what your analysis aims to establish.
3. **Data understanding and preparation.** Summarise the observations used, relevant variables, exploratory findings, cleaning, and feature construction. Explain exclusions and any external resources used.
4. **Methods and justification.** Describe the statistical or machine-learning methods, why they suit your question, and how models or analysis choices were selected. Define any nonstandard quantities.
5. **Results and evaluation.** Present clear tables and figures, appropriate comparisons or baselines, and evidence supporting your claims.
6. **Interpretation and limitations.** Answer your question, distinguish findings from speculation, discuss limitations, and identify useful next steps.
7. **Contributions and references.** Describe each member's concrete contribution and cite data sources, methods, software, pretrained models, and other resources used.

The report will be judged on the following criteria. They describe the 60% report component; they are not additional grading components.

| Criterion | What the assessors will look for |
|---|---|
| Problem formulation | A worthwhile, clearly defined question; convincing motivation; scope suited to the data. |
| Data understanding | Correct interpretation of the fields and observations; careful preparation and informative exploration. |
| Methodological quality | Suitable methods, sound reasoning, justified choices, and appropriate comparisons. |
| Evidence and evaluation | Credible validation or robustness checks; clear results that answer the question; attention to uncertainty and leakage. |
| Interpretation | Conclusions supported by evidence, thoughtful discussion, and recognition of limitations. |
| Communication and reproducibility | Clear organisation, readable slides and figures, complete references, and code that reproduces the reported results. |

For validation, consider whether repeated pairs, images, or cats could be shared across splits. Match the split to your intended generalisation claim and explain its limitations. Fit preprocessing and tune models using the training/validation data, without using withheld test outcomes. A complex model or a high prediction score alone does not guarantee a high report mark.

## 4. Grading, contributions, and the group meeting

The base midterm score uses the following weights:

| Component | Weight | Basis |
|---|---|---|
| Individual contribution | 20% | Your actual contribution to the project. |
| Peer evaluation | 20% | Evaluation of contributions and teamwork. |
| Midterm report | 60% | Quality of the submitted report and its supporting analysis and reproducibility. |
| Base total | 100% | Before any bonus. |
| Bonus | Up to 5 points | Based on the Cat Image Arena selection activity, which has already been completed. |

Let $I$, $P$, and $R$ be scores out of 100 for the three base components, and let $B \in [0, 5]$ be the bonus in points. Then

$$\boxed{\text{Midterm score} = \min\{100,\ 0.20I + 0.20P + 0.60R + B\}.}$$

The bonus adds up to five percentage points to the weighted base score; it is not a fourth base component. For example, a base score of 98 with 5 bonus points gives a final midterm score of 100.

### Individual contribution and peer evaluation

Include a contribution statement in the report specifying each member's tasks and outputs, such as problem formulation, data preparation, modelling, validation, coding, interpretation, writing, or presentation. Contributions should be specific enough to verify from the submitted work.

**Peer evaluation will be conducted after the midterm report submission.** The form, instructions, and deadline will be announced separately. Peer evaluations are confidential, as stated in Lecture 1.

### Presentation at the group meeting

**At least half of the members of each group must present at the meeting,** rounding up for an odd group size. For example, 2 out of 4, 3 out of 5. Each presenter must explain a substantive part of the project. Present directly from the submitted PowerPoint or PDF report and prepare to discuss your question, methods, and results. Presentation time arrangement will be announced separately (each takes around 15 minutes).

## 5. Report and code submission

Submit **one PowerPoint (.ppt or .pptx) or PDF report and the accompanying code package per group** by **12:00 noon on Monday, 2 November 2026 (Hong Kong time)**. The report will be used directly for your presentation. The course submission link will be announced. Identify your group in the filenames, for example `Group_12_Report.pptx` (or `Group_12_Report.pdf`) and `Group_12_Code.zip`.

## 6. Code and reproducibility requirements

Your submission must allow the teaching team to reproduce the results in the report. **Marks will be deducted if the reported results cannot be reproduced from the submitted code and instructions.** AI-assisted tools will be used to check reproducibility.

Include the following in the code package:

- All scripts or notebooks needed for preprocessing, analysis, model fitting, evaluation, and generation of reported tables and figures.
- A README with exact setup and execution instructions, the expected folder structure, and the command or notebook order to run the analysis.
- Software and package versions (for example, an environment file), random seeds, model settings, and any saved split definitions needed to recover the reported results.
- Details of hardware, expected runtime, external downloads, and pretrained weights or other resources when relevant. Use portable relative paths and explain where the supplied dataset should be placed.

Test the instructions in a fresh session before submitting. Identify any remaining numerical variation and its expected size. Saving notebook outputs or screenshots alone is not sufficient: the code must generate the results.

## Permitted dataset use

**By downloading and using the dataset, you agree to use it only for education and research.** Commercial use and use for other purposes are not permitted. Respect the supplied data-use restriction when storing or sharing data, images, and derived materials.